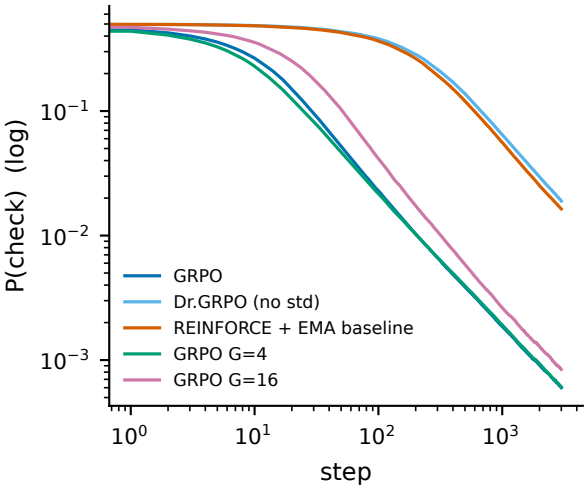
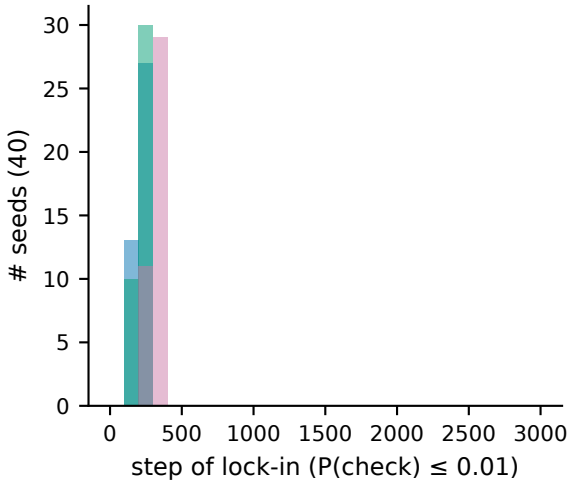

outcome-only, $\pi$=0.9, gap $\pi - (a - c)$ = 0.05
P(check) (log)
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{0}$
$10^{1}$
$10^{2}$
$10^{3}$
step
GRPO
Dr.GRPO (no std)
REINFORCE + EMA baseline
GRPO G=4
GRPO G=16


GRPO's std-normalisation makes lock-in fast
# seeds (40)
0
5
10
15
20
25
30
0
500
1000
1500
2000
2500
3000
step of lock-in (P(check) ≤ 0.01)